Design and Optimisation of an Automotive Rear Window Defroster

MULTIPHYSICS MODELLING USING COMSOL

ME46095

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April 15, 2026

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1 Model Definition

1.1 Introduction

Rear window condensation and frost are a common problem in automotive vehicles, especially during the colder months when the glass surface temperature drops below the dew point of surrounding air [1]. When this happen, the moisture from cabin air will condense on the inner surface of the glass, this significantly reduces the driver's rear visibility and create a safety hazard [2]. To address this issue, most of the modern vehicles are equipped with electrically heated rear window defrosters which uses thin resistive traces that are embedded on the glass to raise it's surface temperature above the dew point.

Problem Statement

The aim of this study is to design and also optimise a resistive rear window defroster system for standard automotive application, using finite element multiphysics modelling in COMSOL Multiphysics 6.2. More specifically, the study seek to answer the following questions:

1. What is the optimal number of heating traces and the operating voltage that is needed to make sure the entire glass surface temperature exceed the dew point?
2. Which trace material out of the available candidates (sourced from CES EduPack database [3]) provides the best balance between thermal performance and structural integrity?
3. How long does it take for the defroster to fully clear the window from a initial frosted condition, and can this be improved by using non-uniform trace spacing?
4. Is the final design robust enough to perform adequately across a range of realistic weather condition?

The defroster must raise the entire glass surface temperature above the dew point to eliminates condensation and frost at the same time it should remaining within safe thermal and mechanical stress limits. The design was developed iteratively through a series of simulations with increasing complexity. We started from a baseline model and then we refining the configuration through parametric sweeps, material selection, stress analysis, trace spacing optimisation, and at the end we also performed a weather study for robustness testing.

1.2 Problem Description

Rear window defrosters work by passing an electric current through thin conductive traces that are deposited on the inner surface of the glass [4]. The Joule heating that is produced in the traces conducts into the glass, which raises it's temperature above the dew point [5]. The main design challenge here is achieving sufficiently uniform heating across the entire window, while using a minimum number of traces at a voltage compatible with the standard $12\,\mathrm{V}$ automotive electrical system [6].
The design must satisfies three main criteria:

1. **Thermal:** The minimum surface temperature of glass must exceeds the dew point temperature ($T_{\text{dew}} = 283.15\,\mathrm{K}$, at $10\,^{\circ}\mathrm{C}$) [1].
2. **Structural:** The peak glass stress must stays below the tempered-glass fracture strength of $165\,\mathrm{MPa}$ [7].
3. **Transient:** The window should be defrosted within a reasonable time (target: $\leq 15\,\mathrm{min}$) [8].

1.3 Geometry IRIAN

The model is represented by a two-dimensional flat rear window, which is a common approach used in literature for this type of problem also [4, 5]. The geometry consists of three components:

- A rectangular glass pane of dimensions $L_{\text{glass}} \times H_{\text{glass}} = 1.5\,\mathrm{m} \times 0.8\,\mathrm{m}$
- Horizontal conductive heating traces spanning the full width of the glass. These are created using a linear array
- Two vertical bus bars (left and right edges) connecting all traces to the power supply

The number of traces (n_{traces}) is a key design variable, swept from 8 to 22 in the parametric studies to find the optimal trace pattern. The traces are uniformly spaced across the glass height. Figure 1 shows the COMSOL geometry for the baseline configuration.

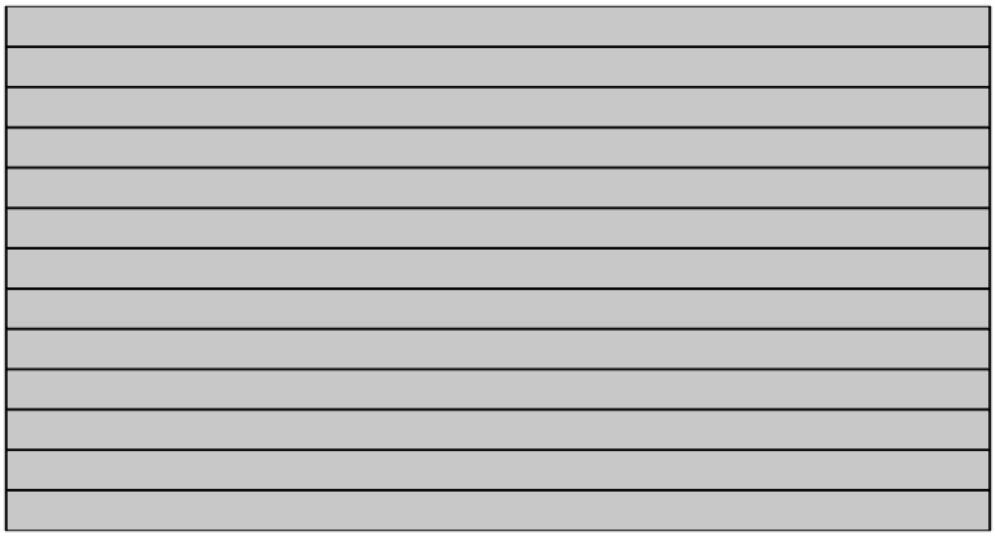


Figure 1: Two-dimensional geometry of the rear window defroster model showing the glass pane, horizontal heating traces, and vertical bus bars. Configuration shown: $n_{\text{traces}} = 12$.

The geometric parameters used in COMSOL are listed in Table 1.

Table 1: Geometric parameters of the rear window defroster model.

Parameter	**Symbol**	**Value**	**Description**
Glass length	L_{glass}	$1.5\,\text{m}$	Width of the window
Glass height	H_{glass}	$0.8\,\text{m}$	Height of the window
Glass thickness	t_{glass}	$5\,\text{mm}$	Out-of-plane thickness
Trace width	w_{trace}	$1.7\,\text{mm}$	Width of each heating trace
Trace thickness	t_{trace}	$0.11\,\text{mm}$	Thickness of each trace
Number of traces	n_{traces}	8–22 (swept)	Design variable

Since the model is two-dimensional with out-of-plane thickness, an effective 2D electrical conductivity is used for the traces:

$$\sigma_{\text{eff}} = \sigma_{\text{trace}} \cdot \frac{t_{\text{trace}}}{t_{\text{glass}}} \tag{1}$$

This accounts for the fact that the thin conductive traces occupy only a fraction of the out-of-plane glass thickness. The trace dimensions used here is consistent with typical screen-printed defroster traces reported in the literature [8].

1.4 Materials

The window glass is modelled as tempered soda-lime glass using COMSOL's built-in material library. The key properties are listed in Table 2.

Table 2: Material properties of tempered soda-lime glass.

Property	Value	Source
Thermal conductivity (k)	$1.4\,\mathrm{W\,m^{-1}\,K^{-1}}$	COMSOL library
Density (ρ)	$2500\,\mathrm{kg\,m^{-3}}$	COMSOL library
Heat capacity (C_p)	$750\,\mathrm{J\,kg^{-1}\,K^{-1}}$	COMSOL library
Fracture strength	$165\,\mathrm{MPa}$	[7]

Four candidate trace materials were evaluated, with properties sourced from the CES EduPack database [3]. The full property set is given in Table 3.

Table 3: Trace material properties from CES EduPack [3]. σ: electrical conductivity, α: coefficient of thermal expansion, E: Young's modulus, ν: Poisson's ratio.

Material	σ ($\mathrm{S\,m^{-1}}$)	α ($\mathrm{K^{-1}}$)	E (GPa)	ν	Yield (MPa)
Pd–Ag alloy	2.3×10^6	15×10^{-6}	102.5	0.38	410
Silver (comm. purity)	5.75×10^7	19.7×10^{-6}	71	0.39	190
Gold (soft annealed)	4.2×10^7	14×10^{-6}	78.5	0.42	20
Nichrome V	9.3×10^5	13×10^{-6}	212.5	0.29	365

1.5 Boundary Conditions IRIAN

The boundary conditions set for the base simulation represent a cold winter morning scenario, where the outside of the glass is exposed to ambient air and the interior is facing the inside of the car. The values are the base values for all simulations and listed in Table 4.

Table 4: Boundary conditions and environmental parameters.

Parameter	Symbol	Value	Description
Ambient temperature	T_{amb}	$275.15\,\mathrm{K}$ ($2\,^\circ\mathrm{C}$)	Exterior air
Cabin temperature	T_{cabin}	$293.15\,\mathrm{K}$ ($20\,^\circ\mathrm{C}$)	Interior cabin air
Initial temperature	T_{init}	$278.15\,\mathrm{K}$ ($5\,^\circ\mathrm{C}$)	Frosted window (transient)
Dew point temperature	T_{dew}	$283.15\,\mathrm{K}$ ($10\,^\circ\mathrm{C}$)	Target minimum temperature
External heat transfer coeff.	h_{ext}	$15\,\mathrm{W\,m^{-2}\,K^{-1}}$	Exterior convection
Internal heat transfer coeff.	h_{int}	$7\,\mathrm{W\,m^{-2}\,K^{-1}}$	Interior convection
Applied voltage	V_{in}	10–$14.5\,\mathrm{V}$ (swept)	Automotive 12V system range

The dew point temperature T_{dew} is estimated from the cabin air conditions using the Magnus approximation [1]:

$$T_{\text{dew}} = \frac{b\,\gamma(T,\mathrm{RH})}{a - \gamma(T,\mathrm{RH})}, \qquad \gamma(T,\mathrm{RH}) = \ln\left(\frac{\mathrm{RH}}{100}\right) + \frac{a\,T}{b+T} \tag{2}$$

with Magnus constants $a = 17.625$ and $b = 243.04\,^\circ\mathrm{C}$, and T in $^\circ\mathrm{C}$. For a cabin at $T_{\text{cabin}} = 20\,^\circ\mathrm{C}$ and a representative relative humidity $\mathrm{RH} = 50\,\%$:

$$\gamma = \ln(0.50) + \frac{17.625 \cdot 20}{243.04 + 20} \approx 0.647 \quad \Rightarrow \quad T_{\text{dew}} \approx \frac{243.04 \cdot 0.647}{17.625 - 0.647} \approx 9.3\,^\circ\mathrm{C}.$$

This is rounded up to $T_{\text{dew}} = 10\,^\circ\mathrm{C}$ ($283.15\,\mathrm{K}$) to provide a small safety margin against higher cabin humidity (e.g. from passengers breathing).
Furthermore a electric potential V_{in} is applied on the left bus bar, and ground ($V = 0$) on the right bus bar. The electric current physics are defined only on the trace and bus bar domains and not on the glass.
For the stress analysis (Sim 2c), fixed constraints are applied on the left and right outer edges of the glass, representing the car body frame which holds the window. This is a simplification of a real window since in an actual window the glass is surrounded by an elastic sealant which provides some complience.

1.6 Physics Coupling and Governing Equations

The model involve three coupled physics interfaces: Electric Currents (ec), Heat Transfer in Solids (ht), and Solid Mechanics (solid). Each of these physics captures a different aspect of the defroster behaviour, and

they are coupled together through multiphysics nodes in COMSOL. The coupling between them is illustrated in Figure 2.
The first physics is **Electric Currents**, which solves for the electric potential distribution across the conductive traces and bus bars. When a voltage is applied across the bus bars, current will flows through the traces and this generates resistive heating. This physics is only active on the trace and bus bar domains since the glass itself is electrically insulating. The second physics is **Heat Transfer in Solids**, which computes the temperature field across the entire glass and trace domains. It recieves the Joule heating from Electric Currents as a volumetric heat source through the *Electromagnetic Heating* multiphysics coupling. This is what actually heats up the glass. The third physics is **Solid Mechanics**, which calculate the mechanical stresses and deformations that arises from the temperature differences in the model. It is coupled to Heat Transfer through the *Thermal Expansion* multiphysics node, which converts temperature changes into thermal strains. This coupling is important because the trace materials and the glass has different coefficients of thermal expansion, so when they heat up together they will expand by different amounts which creates stress at the interface.

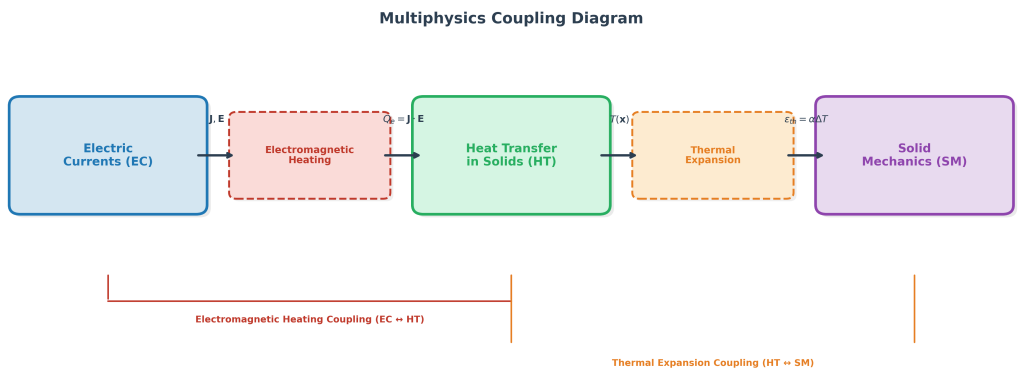


Figure 2: Physics coupling in the multiphysics model. Joule heating from Electric Currents drives Heat Transfer, which in turn drives thermal expansion in Solid Mechanics.

The governing equations for each physics are given below.
Electric Currents. The electric field in the conductive traces are governed by current conservation:

$$\nabla \cdot \mathbf{J} = 0 \tag{3}$$

where $\mathbf{J} = \sigma_{\text{eff}}\mathbf{E}$ is the current density and $\mathbf{E} = -\nabla V$ is the electric field.
Heat Transfer in Solids. The heat equation in the glass domain are:

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q_e + Q_{\text{conv}} \tag{4}$$

where $Q_e = \mathbf{J} \cdot \mathbf{E}$ is the resistive (Joule) heating that comes from the Electromagnetic Heating multiphysics coupling, and Q_{conv} is the convective heat source from **??**.
For stationary studies the left-hand side is zero so the equation reduce to:

$$\nabla \cdot (k \nabla T) + Q_e + Q_{\text{conv}} = 0 \tag{5}$$

Solid Mechanics. The mechanical equilibrium are described by:

$$\nabla \cdot \mathbf{S} + \mathbf{F}_v = 0 \tag{6}$$

where $\mathbf{S}$ is the Cauchy stress tensor. Thermal expansion introduce an initial strain:

$$\varepsilon_{\text{th}} = \alpha(T - T_{\text{ref}})\mathbf{I} \tag{7}$$

with a reference temperature of $T_{\text{ref}} = T_{\text{amb}} = 275.15\,\text{K}$.

2 Model Implementation in COMSOL

The design was developed iteratively through five simulations of increasing complexity. Each simulation builds on the previous one, progressively refining the design from a baseline model to a fully optimised transient solution. Table 5 summarises all simulations.

Table 5: Overview of simulations performed.

Simulation	Study Type	Sweep Parameters	Objective
Sim 1	Stationary	None (baseline)	Verify model physics
Sim 2	Stationary	V_{in}, n_{traces}	Identify viable operating range
Sim 2b	Stationary	Material, n_{traces}, V_{in}	Material optimisation
Sim 2c	Stationary (sequential)	Material, n_{traces}	Stress analysis
Sim 3	Time dependent	Material, n_{traces}, V_{in}	Defrost time
Sim 4	Time dependent	Zone layout (n_{top}, n_{mid}, n_{bot})	Zoned trace spacing
Sim 5	Time dependent	Chebyshev exponent p	Non-uniform spacing optimisation
Sim 6	Time dependent	Weather scenario	Robustness across climate conditions

2.1 Simulation 1 — Baseline Stationary Model

The first simulation establishes the baseline model with the default configuration: 12 traces of silver paste ($\sigma = 2\times10^6\,\mathrm{S\,m^{-1}}$) at $V_{\text{in}} = 12\,\mathrm{V}$. A stationary study couples Electric Currents and Heat Transfer via Electromagnetic Heating.
The purpose of Sim 1 is to verify that the voltage gradient across the traces is correct (12 V at the left bus bar, 0 V at the right), the temperature distribution is physically realistic (in the range 270–320 K) and the cold spots are located at the expected positions (top and bottom edges farthest from traces).

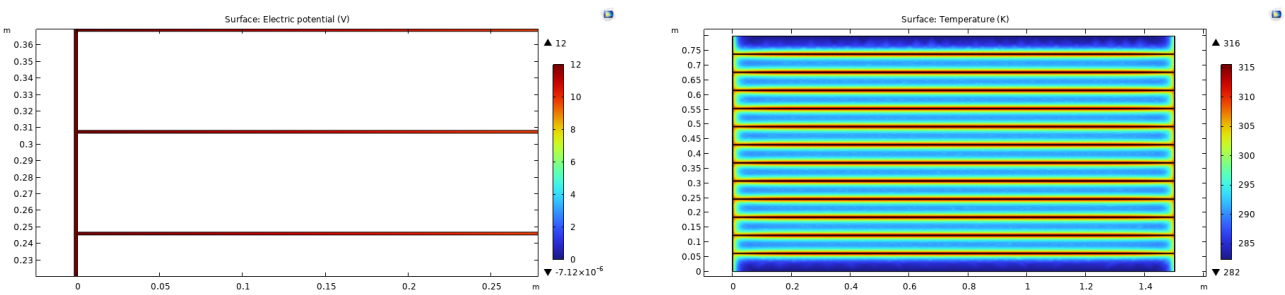


Figure 3: Sim 1 baseline results at $V_{\text{in}} = 12\,\mathrm{V}$, $n_{\text{traces}} = 12$, silver paste. (a) Electric potential distribution on the traces and bus bars. (b) Temperature distribution across the glass surface.

The baseline results show a surface-average temperature of $297.13\,\mathrm{K}$ (above the dew point) but a surface-minimum temperature of $282.12\,\mathrm{K}$ (below the dew point). The cold spots are at the top and bottom edges of the glass where there is no trace coverage, thus indicating room for trace pattern optimization.

2.2 Simulation 2 — Voltage and Trace Count Sweeps

Sim 2 investigates the sensitivity of the minimum glass temperature to the applied voltage and number of traces, independently and in combination.
A parametric sweep of V_{in} from $8\,\mathrm{V}$ to $16\,\mathrm{V}$ (9 values) is performed at $n_{\text{traces}} = 12$. Even at $16\,\mathrm{V}$, the minimum temperature ($283.09\,\mathrm{K}$) barely reaches the dew point, indicating that voltage alone cannot solve the problem with 12 traces.
A parametric sweep of n_{traces} from 8 to 22 is performed at $V_{\text{in}} = 12\,\mathrm{V}$. Results are shown in Figure 4.

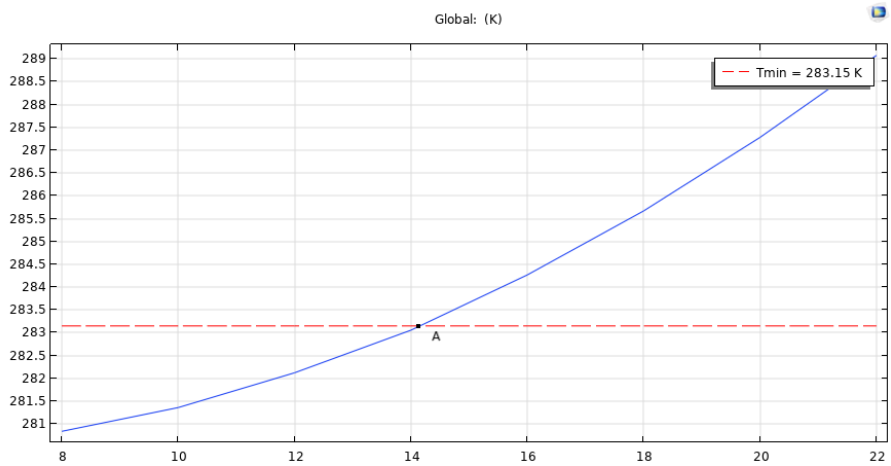


Figure 4: Minimum glass surface temperature as a function of the number of traces at $V_{\text{in}} = 12\,\text{V}$. The horizontal dashed line indicates the dew point temperature ($283.15\,\text{K}$).

A full factorial sweep over $n_{\text{traces}} \in \{8, 10, 12, 14, 16, 18, 20, 22\}$ and $V_{\text{in}} \in \{8, 10, 12, 14, 16\}$ V produces 40 stationary solutions. Results are shown in Table 6.

Table 6: Minimum surface temperature T_{min} (K) for the combined voltage–trace count sweep. Values $\geq 283.15\,\text{K}$ are shown in **bold**.

n_{traces}	$8\,\text{V}$	$10\,\text{V}$	$12\,\text{V}$	$14\,\text{V}$	$16\,\text{V}$
8	280.86	280.85	280.84	280.83	280.81
10	281.09	281.21	281.36	281.54	281.74
12	281.43	281.74	282.12	282.57	**283.09**
14	281.85	282.39	283.06	**283.84**	**284.75**
16	282.38	**283.23**	**284.27**	**285.49**	**286.91**
18	283.01	**284.21**	**285.67**	**287.40**	**289.40**
20	**283.72**	**285.33**	**287.28**	**289.60**	**292.27**
22	**284.52**	**286.57**	**289.08**	**292.04**	**295.46**

The table concludes that increasing the number of traces is more effective than increasing voltage. The cheapest solution above the dew point is 16 traces at $10\,\text{V}$ ($283.23\,\text{K}$).

2.3 Simulation 2b — Material Optimisation

Sim 2b extend the parametric sweep to includes four candidate trace materials from the CES EduPack database [3]. The sweep is covering $4 \times 8 \times 4 = 128$ combinations (4 materials, 8 trace counts and 4 voltages).
The voltages used reflects the realistic automotive range: $10\,\text{V}$ (battery low), $12\,\text{V}$ (nominal), $13.5\,\text{V}$ (engine is running), and $14.5\,\text{V}$ (maximum alternator output).
The minimum number of traces which is needed to reach the dew point for each material and voltage are shown in Table 7.

Table 7: Minimum number of traces to reach $T_{\text{min}} \geq 283.15\,\text{K}$ for each material and voltage.

Material	$10\,\text{V}$	$12\,\text{V}$	$13.5\,\text{V}$	$14.5\,\text{V}$
Pd–Ag alloy	16	14	14	14
Nichrome V	22	18	18	16
Silver (comm. purity)	*Overheats — not viable*			
Gold (soft annealed)	*Overheats — not viable*			

For Pd–Ag alloy, the dew point was been reached at as few as 14 traces at $12\,\text{V}$ ($283.38\,\text{K}$), going all the way up to $294.65\,\text{K}$ at 22 traces and $14.5\,\text{V}$, which gives a quite wide viable operating range. Nichrome V on the other hand need more traces to achieves the same result — the dew point was first reach at 16 traces and $14.5\,\text{V}$

($283.18\,\mathrm{K}$), and the maximum T_{min} that was recorded across the whole sweep is only $286.45\,\mathrm{K}$ at 22 traces and $14.5\,\mathrm{V}$. The full trends for both of the materials can be seen in Figure 5.

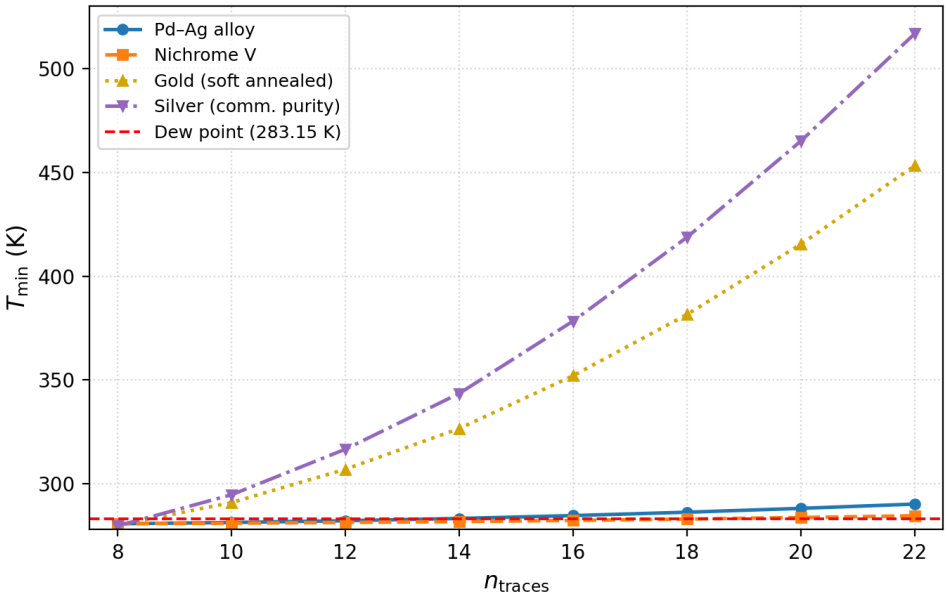


Figure 5: Comparison of minimum surface temperature vs. number of traces for all four EduPack materials at $V_{\text{in}} = 12\,\mathrm{V}$. Silver and Gold exceed safe temperatures at $n_{\text{traces}} \geq 10$ and are not viable.

- **Silver (commercial purity) and Gold (soft annealed) are not viable** — their bulk conductivity ($> 10^7$ S/m) are too high for the screen-printed geometry, causing extreme overheating (Silver reach $625\,\mathrm{K}$ at 22 traces, $14.5\,\mathrm{V}$).
- **Pd–Ag alloy is the best EduPack material** — it reaches the dew point at 14 traces ($12\,\mathrm{V}$+). The temperatures remains in a safe range.
- **Nichrome V is viable but requires more traces** (18–22 depending on the voltage).

2.4 Simulation 2c — Stress Analysis

Sim 2c evaluate the thermal stresses for the two shortlisted materials (Pd–Ag alloy and Nichrome V). Silver and Gold was excluded from this analysis as both materials overheats severely in Sim 2b, making them not viable candidate for the defroster design. Since stress analysis are only relevant for materials that actually meets the thermal criteria, there is no point to included them here.
Because the geometry was built with Form Union, the reported peak von Mises stress is interpreted here as the stress in the glass body, not as a seperate trace-only stress by it self. For that reason the structural criterion used in this section is then $\sigma_{\text{max}} < 165\,\mathrm{MPa}$ for tempered glass [7].
The voltage range were also narrowed down to $12\,\mathrm{V}$ and $14.5\,\mathrm{V}$ only. The $12\,\mathrm{V}$ represents the nominal battery voltage which is the primary operating condition, and $14.5\,\mathrm{V}$ represents the maximum alternator output which is the most worst-case scenario in terms of Joule heating and therefore thermal stress. The intermediate voltages of $10\,\mathrm{V}$ and $13.5\,\mathrm{V}$ was not included as they do not adds additional informations beyond what these two extreme cases already captures.
For the trace count, only configurations from $n_{\text{traces}} = 14$ to $n_{\text{traces}} = 22$ was selected. This is because Sim 2b showed that below 14 traces, neither material is able to reaches the dew point at any of the tested voltages, so performing a stress analysis on these configurations would of been pointless.
Solid Mechanics and Thermal Expansion are re-enabled, and a sequential solve were used:

- **Step 1:** Electric Currents + Heat Transfer (stationary)
- **Step 2:** Solid Mechanics with Thermal Expansion (stationary, using Step 1 temperatures)

This sequential approach avoid the spatial frame conflict ("dvol_spatial undefined") which is encountered when solving all the physics simultaneously in COMSOL 6.2.

Table 8: Maximum glass von Mises stress for designs with Pd–Ag traces. Values above the $165\,\mathrm{MPa}$ glass limit is shown in **bold**.

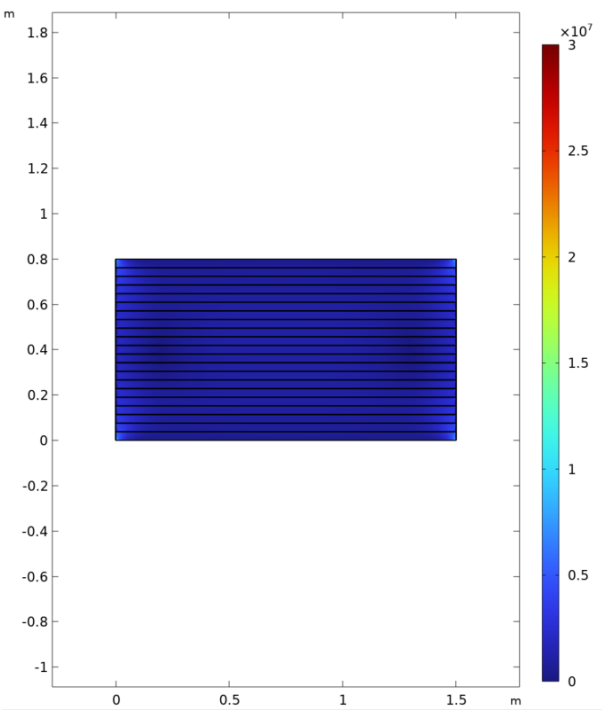
n_{traces}	**Max glass stress at $12\,\mathrm{V}$ (MPa)**	**Max glass stress at $14.5\,\mathrm{V}$ (MPa)**
14	120.1	**167.7**
16	126.5	**176.9**
18	133.4	**187.0**
20	141.2	**198.4**
22	150.9	**212.6**

Table 9: Maximum glass von Mises stress for designs with Nichrome V traces for the tested cases.

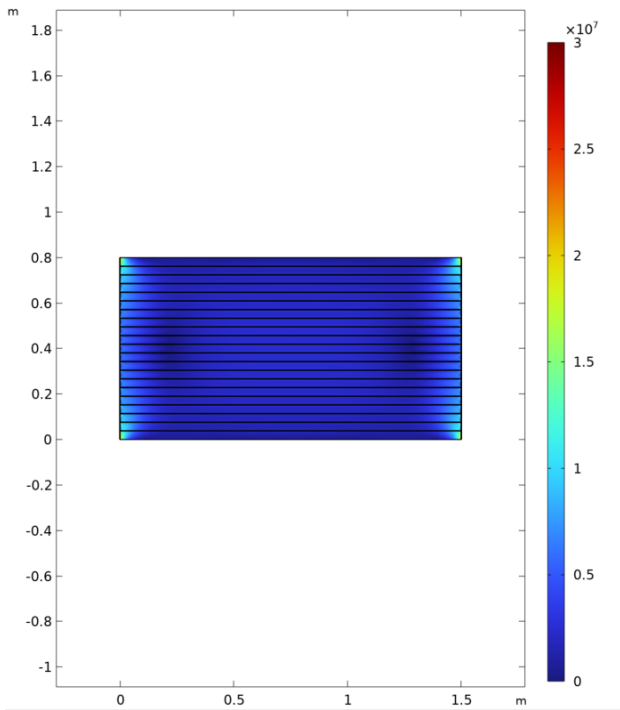
n_{traces}	**Max glass stress at $12\,\mathrm{V}$ (MPa)**	**Max glass stress at $14.5\,\mathrm{V}$ (MPa)**
14	93.9	124.9
16	98.2	131.0
18	102.6	137.6
20	106.7	143.7
22	111.2	150.2

Table 10: Configurations that satisfies both the thermal criterion from Sim 2b and the glass-stress criterion from Sim 2c.

Material	**Voltage**	**Trace counts that pass**
Pd–Ag alloy	$12\,\mathrm{V}$	14, 16, 18, 20, 22
Pd–Ag alloy	$14.5\,\mathrm{V}$	none
Nichrome V	$12\,\mathrm{V}$	18, 20, 22
Nichrome V	$14.5\,\mathrm{V}$	16, 18, 20, 22



(a) Nichrome V (max glass stress = $106.7\,\mathrm{MPa}$, in the glass domain)



(b) Pd–Ag alloy (max glass stress = $141.2\,\mathrm{MPa}$, in the glass domain)

Figure 6: Glass von Mises stress distribution at $V_{\text{in}} = 12\,\mathrm{V}$, $n_{\text{traces}} = 20$ for both trace materials. The highest stresses are at the trace ends near the fixed side edges, so these peaks is conservative because a real rear window is held by rubber and can move a bit.

The stress comes from the thermal expansion mismatch between the conductive traces and the soda-lime glass. Pd–Ag has a higher expansion coefficient than the glass, so when it heats up it tries to expand more and this loads the glass close to the trace ends. Because of the Form Union setup, the peak values in Tables 8 and 9 are treated as glass stresses and not only the traces.
Using the tempered-glass limit of $165\,\mathrm{MPa}$ [7], all Pd–Ag cases at $12\,\mathrm{V}$ pass the structural criterion ($120.1\,\mathrm{MPa}$ to $150.9\,\mathrm{MPa}$), but every Pd–Ag case at $14.5\,\mathrm{V}$ fails it ($167.7\,\mathrm{MPa}$ to $212.6\,\mathrm{MPa}$). Nichrome V stays below the glass limit for all tested stress cases, but only the combinations listed in Table 10 are good overall because they also meets the thermal requirement too. So the higher $14.5\,\mathrm{V}$ option is not acceptable anymore for the preferred Pd–Ag design then.
The highest stresses appears at the left and right edges where the traces end and where the model is fully fixed. In real life these window edges are not perfectly fixed, there is rubber and sealant present, so the glass can move a little bit around. That means the predicted edge peak is likely conservative, but for design selection it is still better to keep the safer $12\,\mathrm{V}$ cases anyway.

2.5 Simulation 3 — Transient Defrost Study

Sim 3 evaluates how long each configuration take to defrost the window starting from the initial frosted condition ($T_{\text{init}} = 278.15\,\mathrm{K}$) until the point where $T_{\text{min}} \geq T_{\text{dew}}$.
A time-dependent study were used with output times from 0 to $3600\,\mathrm{s}$ in $60\,\mathrm{s}$ increments. The Electric Currents equation was set to stationary form since the electric field equilibrate in microseconds compared to the thermal time scale which is in minutes. Solid Mechanics was disabled for computational efficiency since we do not need the stress results for this simulation.
The sweep cover: 2 materials $\times$ 5 trace counts (14, 16, 18, 20, 22) $\times$ 2 voltages (12, $14.5\,\mathrm{V}$) = 20 runs. The mesh were also refined to Fine (189,192 domain elements) for better accuracy.
The overall defrost-time comparison for both materials is shown in Figure 7, while the detailed transient response for Pd–Ag alloy at $12\,\mathrm{V}$ is shown in Figure 8.

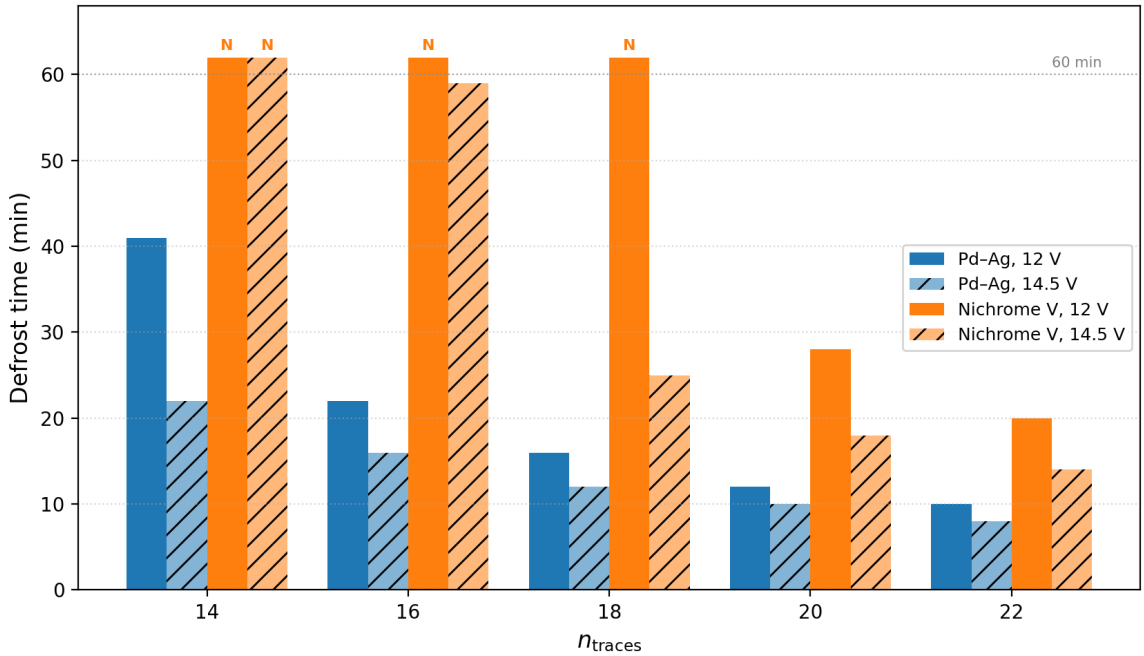


Figure 7: Defrost time comparison between Pd–Ag alloy and Nichrome V at $12\,\mathrm{V}$ and $14.5\,\mathrm{V}$. “N” indicates the configuration never reaches the dew point within $60\,\mathrm{min}$.

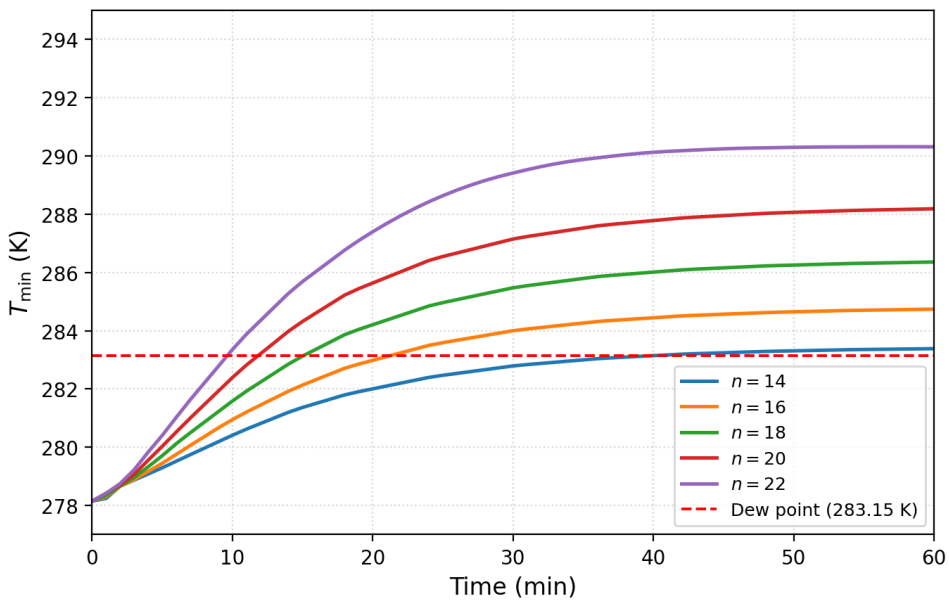


Figure 8: Transient evolution of minimum surface temperature for Pd–Ag alloy at $V_{\text{in}} = 12\,\mathrm{V}$ for different numbers of traces. The horizontal dashed line indicates the dew point.

Pd–Ag alloy defrosts faster than Nichrome V across the full sweep. At $12\,\mathrm{V}$, Pd–Ag reaches the dew point with as few as 14 traces, although slowly $(41\,\mathrm{min})$, and improves to $12\,\mathrm{min}$ at 20 traces and $10\,\mathrm{min}$ at 22 traces. Nichrome V performs noticeably worse: it does not defrost at all below 20 traces at $12\,\mathrm{V}$, and even at $14.5\,\mathrm{V}$ it only reaches the dew point in $59\,\mathrm{min}$ for 16 traces. This confirms that Pd–Ag alloy provides the more practical operating window for battery-level voltage.
Based on these results, **Pd–Ag alloy with $n_{\text{traces}} = 20$ at $V_{\text{in}} = 12\,\mathrm{V}$ was selected as the recommended design**. The key reasons for this was:

- **Stress compliance:** The Pd–Ag cases at $14.5\,\mathrm{V}$ all goes above the $165\,\mathrm{MPa}$ glass limit in Sim 2c, so that higher-voltage option had to be removed even though it defrosts faster still.
- **Battery-only operation:** A nominal automotive 12 V system operates directly from the battery without

the engine running [6]. This matters for cold starts, where the rear defroster should already work before the alternator is active yet.

- **Acceptable defrost time:** At $12\,\mathrm{V}$, 20 traces defrosts the window in $12\,\mathrm{min}$, which is practical enough for the intended use for this design.
- **Adequate thermal margin:** The steady-state $T_{\text{min}} = 288.2\,\mathrm{K}$ is $5.05\,\mathrm{K}$ above the dew point, so there is still some margin against condensation coming back again.
- **Better balance than 22 traces:** Going to 22 traces only saves 2 minutes at $12\,\mathrm{V}$, while using more material and also pushing the glass stress closer to the limit ($150.9\,\mathrm{MPa}$ vs. $141.2\,\mathrm{MPa}$), which is not really worth it anymore.

At first the $14.5\,\mathrm{V}$ Pd–Ag cases looked attractive because they defrost quicker, but after re-reading the stress output they does not comply with the glass-stress criterion. For that reason they were not carried forward as an alternative design anymore.

2.6 Simulation 4 — Trace Pattern Optimisation

After establishing the optimal material (Pd–Ag alloy) and operating conditions in the previous simulations, Sim 4 investigates the effect of trace layout on the temperature uniformity across the glass surface. The baseline simulations revealed that cold spots form at the top and bottom edges of the glass, farthest from any heating trace. A natural mitigation strategy is to concentrate more traces near these edges.
The glass is divided into three horizontal zones with independent trace densities:

- **Top zone:** $H_{\text{top}} = 200\,\mathrm{mm}$, $n_{\text{top}} = 6$ traces
- **Middle zone:** $H_{\text{mid}} = 400\,\mathrm{mm}$, $n_{\text{mid}} = 8$ traces
- **Bottom zone:** $H_{\text{bot}} = 200\,\mathrm{mm}$, $n_{\text{bot}} = 6$ traces

This gives a total of 20 traces (same as the recommended uniform design) but with denser spacing in the edge zones. The edge gap is $28.6\,\mathrm{mm}$ (vs. $38.1\,\mathrm{mm}$ uniform), and the centre gap is $44.4\,\mathrm{mm}$.
Zone parameters were added to Global Definitions and each zone was created as a separate Rectangle + Array in the geometry.

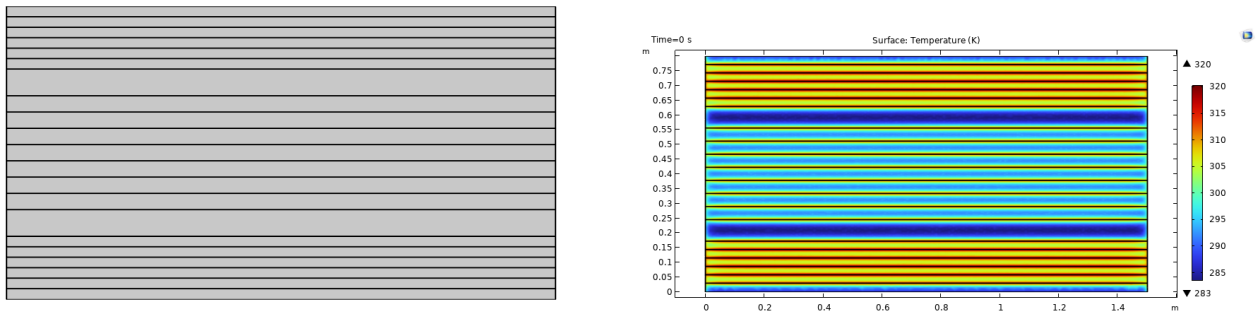


Figure 9: Zoned trace layout (Approach A). (a) Geometry with 6/8/6 distribution across top/middle/bottom zones. (b) Temperature distribution at $T = T_{\text{dew}}$ showing reduced cold spots at the edges.

The study was changed from stationary to time dependent with output range `range(0,60,3600)`, i.e. every $60\,\mathrm{s}$ for $60\,\mathrm{min}$. The Electric Currents equation form was set to Stationary (as in Sim 3) since the electric field equilibrates instantaneously compared to the thermal time scale. The initial temperature was set to $T_{\text{init}} = 278.15\,\mathrm{K}$.
The zoned approach was not pursued further because the continuous Chebyshev spacing (Sim 5) proved far more effective, as discussed in Section 2.7.

2.7 Simulation 5 — Chebyshev Trace Spacing Optimisation

While the zoned approach (Sim 4) provides a discrete improvement, a mathematically elegant alternative is to use a continuous spacing function that naturally clusters traces near the edges. Chebyshev node spacing, derived from the roots of Chebyshev polynomials, is well known in numerical analysis for minimising interpolation error by placing points near interval boundaries. This property is analogous to the defroster design goal of concentrating heating where it is most needed.

Spacing Formula and Implementation

The vertical position y_i of the i-th trace ($i = 1, \ldots, 20$) is given by:

$$y_i = \frac{H_{\text{glass}}}{2} \left(1 - \text{sign}\left(\cos \frac{i\pi}{21} \right) \left| \cos \frac{i\pi}{21} \right|^p \right) \tag{8}$$

where p is a clustering exponent that controls how aggressively the traces are pushed toward the edges:

- $p < 1$: More uniform than standard Chebyshev (less edge clustering)
- $p = 1$: Standard Chebyshev distribution
- $p > 1$: More traces at the edges and in the middle

The $\text{sign}(\cdot)|\cdot|^p$ formulation preserves the symmetry of the cosine distribution for all values of p, including non-integer exponents.
Twenty individual traces were created in the geometry, each with a unique y-position computed from Equation (8). The exponent p was defined as a global parameter, enabling a parametric sweep. All other model settings (material: Pd–Ag alloy, σ_{eff} correction, time-dependent study, boundary conditions) were identical to Sim 4.

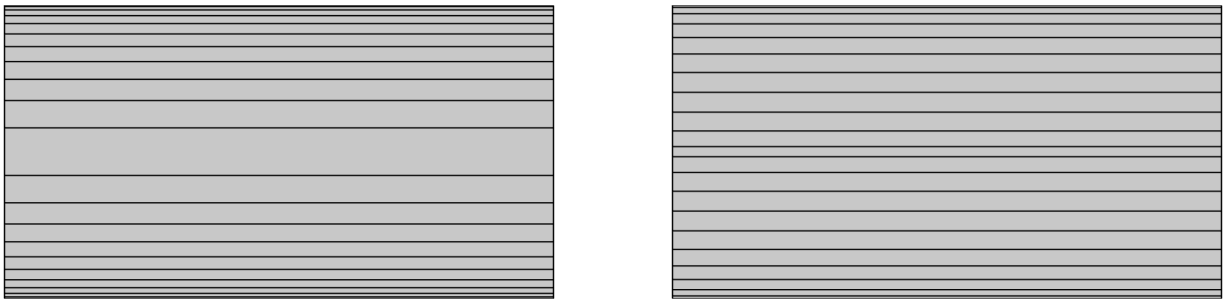


Figure 10: Chebyshev trace spacing for (a) $p = 0.7$ (near-uniform) and (b) $p = 1.3$ (optimal). Edge gaps decrease significantly with increasing p while centre gaps widen.

Results and Optimum

Three values of p were tested first to establish the feasibility of the approach:

- $p = 1.0$ (standard Chebyshev): **defrost in** $7\,\text{min}$, steady-state $T_{\text{min}} = 292.6\,\text{K}$
- $p = 0.85$: **never reaches dew point** — steady-state $T_{\text{min}} = 282.06\,\text{K}$
- $p = 0.7$: **never reaches dew point** — steady-state $T_{\text{min}} = 282.06\,\text{K}$

The $p = 1.0$ shows that the defrost time is reduced from $12\,\text{min}$ (uniform) to $7\,\text{min}$ — a 42% improvement with exactly the same number of traces, same material and same voltage. However, reducing p below 1.0 removes too much edge clustering and the cold spots return.
To find the optimal exponent, a parametric sweep was performed over $p \in \{0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.$
All runs use 20 traces of Pd–Ag alloy at $V_{\text{in}} = 12\,\text{V}$. The trace y-positions were expressed parametrically in terms of p, so COMSOL automatically updates the geometry for each sweep value.
The results are plotted in Figure 11.

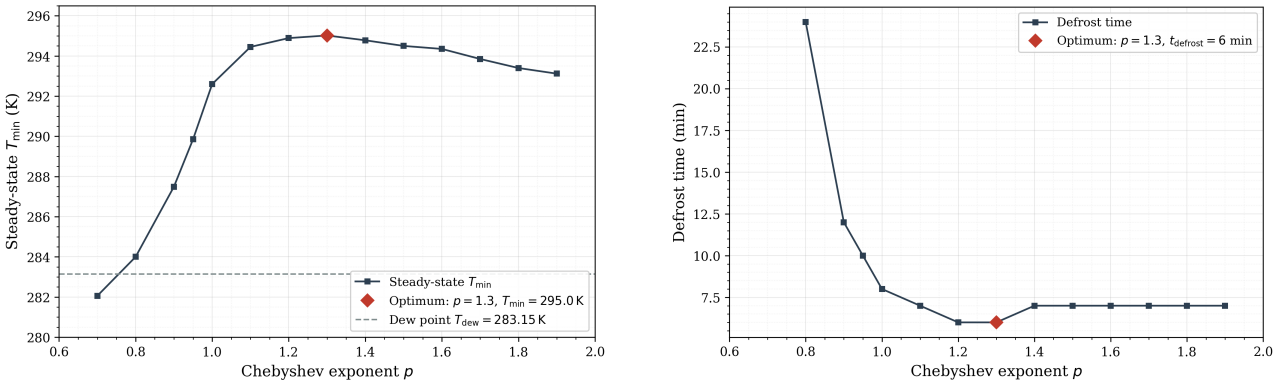


Figure 11: Effect of the Chebyshev exponent p on (a) steady-state minimum temperature and (b) defrost time. The optimum is $p = 1.3$, achieving the highest T_{min} and fastest defrost.

The optimum is clearly at $p = 1.3$: it achieves the highest minimum temperature at the 6-minute mark ($283.20\,\text{K}$) and the highest steady-state T_{min} ($295.0\,\text{K}$). The actual temperature distribution can be visualised in Figure 12, showing a much more uniform temperature across the glass with no cold spots at the edgesor the middle. Hier kan ik gecommente info nog erbij zetten

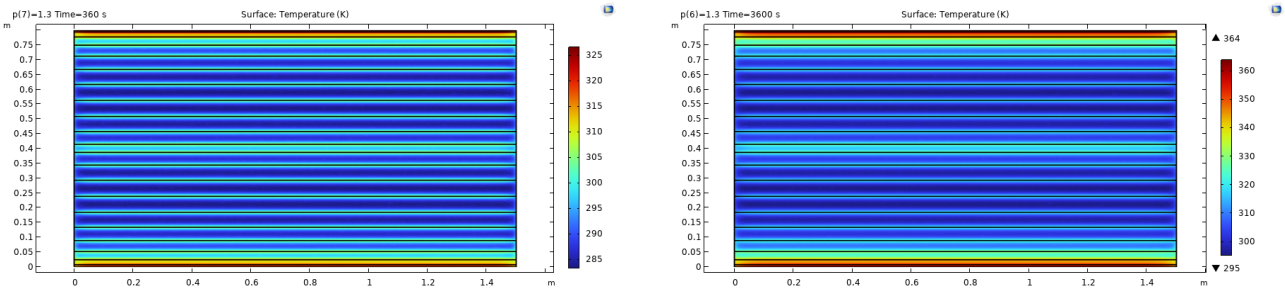


Figure 12: Temperature distribution for Chebyshev spacing $p = 1.3$ (Pd–Ag alloy, 20 traces, $12\,\text{V}$). (a) At $t = 6\,\text{min}$ (defrost moment): $T_{\text{min}} = 283.20\,\text{K}$. (b) Steady state: $T_{\text{min}} = 295.0\,\text{K}$.

Practical Considerations

An important practical constraint for rear window trace patterns is driver visibility. Per the SAE J216 standard, obscuration of the rear vision area should not exceed 5% in the primary zone [2]. For 20 traces with $w_{\text{trace}} = 1.7\,\text{mm}$:

$$\text{Obscuration} = \frac{n_{\text{traces}} \times w_{\text{trace}}}{H_{\text{glass}}} = \frac{20 \times 1.7}{800} = 4.25\% \tag{9}$$

This is well within the 5% limit regardless of the spacing pattern, since the total trace area is unchanged. The Chebyshev spacing only redistributes the same number of traces, so visibility is not affected.
Adding vertical traces between the horizontal ones was considered to create a grid pattern. However, analysis showed this would be ineffective: since all parallel horizontal traces are at the same voltage at any given x-position, vertical connectors between them would have zero potential difference ($\Delta V = 0$) and carry no current. Variable-width traces could create voltage differences enabling vertical current flow, but this adds significant manufacturing complexity without clear benefit over optimising the horizontal spacing alone.

2.8 Simulation 6 — Weather Condition Robustness Study

All previous simulations used a fixed ambient condition: $T_{\text{amb}} = 2\,°\text{C}$, $h_{\text{ext}} = 15\,\text{W}\,\text{m}^{-2}\,\text{K}^{-1}$, and $T_{\text{dew}} = 10\,°\text{C}$. Sim 6 tests the final design (Pd–Ag alloy, 20 traces, Chebyshev $p = 1.3$, $V_{\text{in}} = 12\,\text{V}$) across five realistic weather scenarios to verify that it performs robustly under a range of real-world conditions.

Weather Scenarios

The five scenarios were selected to cover the full range of conditions a vehicle might encounter in Northwestern Europe [9], from mild autumn fog to extreme winter frost. Each scenario defines a physically consistent set

of T_{amb}, T_{dew}, and h_{ext} values — the dew point is not fixed but varies with ambient temperature and relative humidity [1, 10] — as summarised in Table 11.

Table 11: Weather scenarios tested in Sim 6. Each scenario uses a physically realistic dew point consistent with the ambient humidity conditions.

#	Scenario	T_{amb} (°C)	T_{dew} (°C)	h_{ext} ($\mathrm{W\,m^{-2}\,K^{-1}}$)
1	Mild winter	2	10	15
2	Cold winter	-5	0	25
3	Extreme cold	-20	-15	30
4	Cool autumn	10	13	10
5	Freezing rain	-2	0	40

Results

The transient T_{min} for each scenario is shown in Figure 13. Each solid curve is paired with a dashed horizontal line at its scenario-specific dew point. Defrost is achieved when the curve crosses its corresponding dashed line.

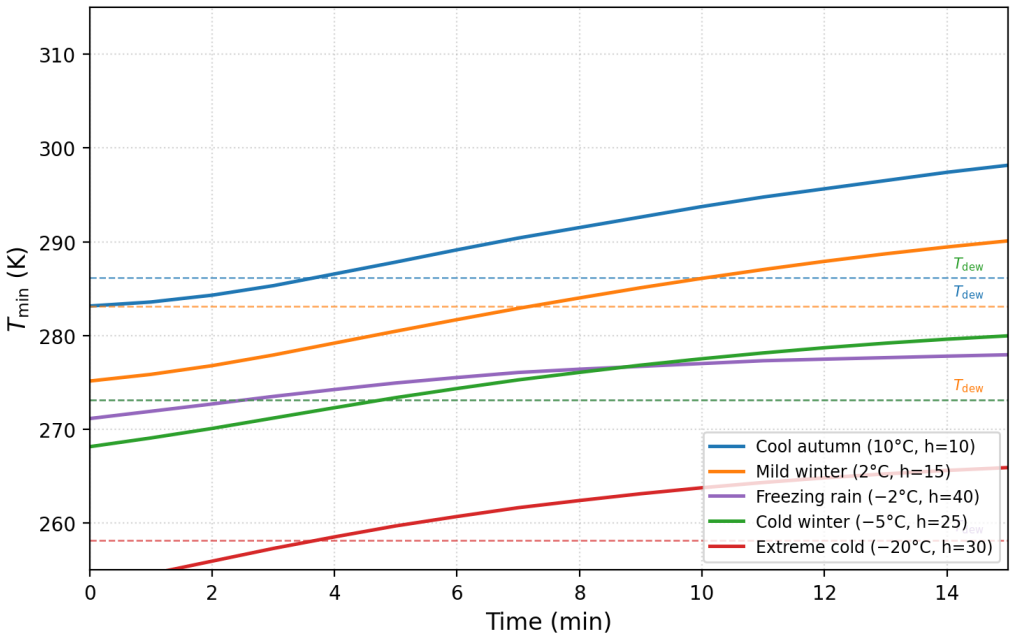


Figure 13: Transient minimum surface temperature for all five weather scenarios (Pd–Ag alloy, 20 traces, Chebyshev $p = 1.3$, $V_{\text{in}} = 12\,\mathrm{V}$). Each dashed line shows the scenario-specific dew point. All curves cross their dew point within $8\,\mathrm{min}$.

A summary of defrost times, steady-state minimum temperatures, and margins above the dew point is given in Table 12.

Table 12: Sim 6 results for all five weather scenarios. Margin $= T_{\text{min,SS}} - T_{\text{dew}}$.

#	Scenario	Defrost time	$T_{\text{min,SS}}$ (K)	T_{dew} (K)	Margin (K)
1	Mild winter	7 min	295.0	283.15	$+11.9$
2	Cold winter	4 min 47 s	281.6	273.15	$+8.5$
3	Extreme cold	3 min 41 s	267.0	258.15	$+8.9$
4	Cool autumn	3 min 40 s	307.5	286.15	$+21.4$
5	Freezing rain	2 min 33 s	278.6	273.15	$+5.4$

All five scenarios pass the defrost criterion with positive margin. Key observations are:

- **Faster defrost under harsher convection:** Scenarios with high h_{ext} (freezing rain: 40, extreme cold: 30) defrost in under 4 minutes. The high convective coefficient drives a steeper thermal gradient, which accelerates the initial transient even though the steady-state temperature is lower.
- **Tightest margin in freezing rain (Scenario 5):** With $h_{\text{ext}} = 40\,\mathrm{W\,m^{-2}\,K^{-1}}$ and a high convective heat loss, the steady-state T_{min} is only $5.4\,\mathrm{K}$ above the dew point. This is the worst-case scenario for sustained condensation prevention.
- **Largest margin in cool autumn (Scenario 4):** Low convective loss and mild ambient conditions result in a $21.4\,\mathrm{K}$ margin — more than sufficient.
- **Robustness across $\Delta T_{\text{amb}} = 30\,\mathrm{K}$:** The design passes all scenarios spanning a $30\,\mathrm{K}$ ambient temperature range ($-20\,^\circ\mathrm{C}$ to $10\,^\circ\mathrm{C}$), demonstrating robustness to real-world variability.

3 Model Validation

3.1 Mesh Convergence Study

A mesh convergence study was performed to verify that all results are independent of mesh resolution. The study was carried out using a stationary Electric Currents and Heat Transfer solve under the Sim 6 Scenario 1 conditions (mild winter: $T_{\text{amb}} = 2\,^\circ\mathrm{C}$, $h_{\text{ext}} = 15\,\mathrm{W\,m^{-2}\,K^{-1}}$), with the final design (Pd–Ag alloy, 20 traces, Chebyshev $p = 1.3$, $V_{\text{in}} = 12\,\mathrm{V}$). Four physics-controlled mesh levels were tested, from Extra Coarse to Extra Fine.

Table 13: Mesh convergence results (stationary EC + HT, Pd–Ag alloy, 20 traces, Chebyshev $p = 1.3$, $V_{\text{in}} = 12\,\mathrm{V}$, Scenario 1 conditions).

Mesh level	**Domain elements**	T_{min} **(K)**	ΔT_{min} **vs Extra Fine**
Extra Coarse	185,852	281.607	0.004 K
Coarse	241,089	281.609	0.002 K
Fine	274,402	281.611	<0.001 K
Extra Fine	319,050	281.611	—

The minimum surface temperature varies by only $0.004\,\mathrm{K}$ across all four mesh levels, confirming that the solution is fully converged even at Extra Coarse resolution for this geometry. The Fine mesh (274,402 elements) is selected for all simulations as it matches the Extra Fine result to within $0.001\,\mathrm{K}$ while keeping solve times practical.
It should be noted that mesh resolution is more critical for configurations where the result falls close to a threshold. For the Nichrome V material near the dew point:

- **Nichrome V, 18 traces, $12\,\mathrm{V}$:** A coarse mesh predicted a defrost time of 48 minutes, while the Fine mesh reveals that the steady-state $T_{\text{min}} = 283.09\,\mathrm{K}$ remains *below* the dew point — the window never fully defrosts.
- **Nichrome V, 16 traces, $14.5\,\mathrm{V}$:** The coarse mesh predicted "Never", while the Fine mesh gives 59 minutes ($T_{\text{min}} = 283.152\,\mathrm{K}$, barely above the dew point).

These borderline cases confirm that the Fine mesh is necessary for reliable results near the dew point threshold, and justify its use throughout this study.

3.2 Comparison to Literature

The model physics and boundary condition values are validated against published experimental and numerical studies.
The convection coefficients used in this model ($h_{\text{ext}} = 15\,\mathrm{W\,m^{-2}\,K^{-1}}$, $h_{\text{int}} = 7\,\mathrm{W\,m^{-2}\,K^{-1}}$) are compared with experimental measurements by Hożejowska et al. [9], who reported measured h values for automotive glass under various driving conditions. Their results support the order of magnitude used in this study, with exterior values in the range $10\,\mathrm{W\,m^{-2}\,K^{-1}}$ to $25\,\mathrm{W\,m^{-2}\,K^{-1}}$ depending on vehicle speed.
Rühl et al. [4] performed coupled finite element simulations of electrically heated glass panes with resistive lines, validated against infrared thermography and thermocouple measurements. Their modelling approach —

coupling electric currents with heat transfer via Joule heating in a 2D representation — is directly analogous to the physics setup used in this work.
Koshlak et al. [5] measured that 83–85% of the heat generated in electrically heated automotive glass is directed inward (toward the cabin side), which is consistent with the asymmetric convection coefficients used in this model ($h_{\text{int}} < h_{\text{ext}}$, meaning less heat loss to the interior).

3.3 Back-of-Envelope Power and Temperature Estimate

An independent analytical estimate of the steady-state average temperature provide a direct check on the numerical results. In this section I used the recommended design from the previous simulations, which is Pd–Ag alloy with 20 traces at $12\,\mathrm{V}$, because this was the final design that got selected. The final Chebyshev spacing changes where the heat is more concentrated, it does not change the total electrical power input, so the same average-temperature estimate can still be used for the final 20-trace design also.
Each trace has electrical resistance:

$$R_{\text{trace}} = \frac{L_{\text{glass}}}{\sigma_{\text{trace}} \cdot w_{\text{trace}} \cdot t_{\text{trace}}} \tag{10}$$

For the selected Pd–Ag design ($\sigma = 2.3 \times 10^6\,\mathrm{S\,m^{-1}}$):

$$R_{\text{trace}} = \frac{1.5}{2.3 \times 10^6 \times 1.7 \times 10^{-3} \times 0.11 \times 10^{-3}} = 3.49\,\Omega \tag{11}$$

The power per trace is $P_{\text{trace}} = V_{\text{in}}^2/R_{\text{trace}} = 144/3.49 = 41.3\,\mathrm{W}$, and the total power for 20 traces is:

$$P_{\text{total}} = 20 \times 41.3 = 825.8\,\mathrm{W} \tag{12}$$

Note that this physical power is correctly reproduced in the 2D COMSOL model through the σ_{eff} correction (Equation (1)): the effective resistance in the model is $R_{\text{model}} = L_{\text{glass}}/(\sigma_{\text{eff}} \cdot w_{\text{trace}} \cdot t_{\text{glass}}) = L_{\text{glass}}/(\sigma_{\text{trace}} \cdot w_{\text{trace}} \cdot t_{\text{trace}}) = R_{\text{trace}}$.
At steady state, the total Joule heating must equal the convective losses from both sides of the glass:

$$P_{\text{total}} = (h_{\text{ext}} + h_{\text{int}}) \cdot A_{\text{glass}} \cdot (T_{\text{avg}} - T_{\text{weighted}}) \tag{13}$$

where $A_{\text{glass}} = L_{\text{glass}} \times H_{\text{glass}} = 1.2\,\mathrm{m}^2$ and the convection-weighted ambient temperature is:

$$T_{\text{weighted}} = \frac{h_{\text{ext}} \cdot T_{\text{amb}} + h_{\text{int}} \cdot T_{\text{cabin}}}{h_{\text{ext}} + h_{\text{int}}} = \frac{15 \times 275.15 + 7 \times 293.15}{22} = 280.9\,\mathrm{K} \tag{14}$$

Solving for T_{avg}:

$$T_{\text{avg}} = T_{\text{weighted}} + \frac{P_{\text{total}}}{(h_{\text{ext}} + h_{\text{int}}) \cdot A_{\text{glass}}} = 280.9 + \frac{825.8}{22 \times 1.2} = 280.9 + 31.3 = 312.2\,\mathrm{K} \tag{15}$$

The analytical estimate for the selected 20-trace Pd–Ag design is therefore a steady-state surface-average temperature of approximately $312.2\,\mathrm{K}$. This basically means the system is going toward this temperature in the long-time limit, not that the whole glass suddenly become this value at one exact time during the early transient. This value is intentionally an *average* over the whole glass area, so it is expected to be much higher than the reported minimum temperatures for the same design (for example $288.2\,\mathrm{K}$ for the uniform 20-trace case, or $295.0\,\mathrm{K}$ for the Chebyshev $p = 1.3$ layout), because those minima happens at the coldest local points near the glass edges and not over the full surface.
The exported COMSOL surface-average data for the final Chebyshev $p = 1.3$ design confirm this estimate pretty good. The average temperature rises from $278.15\,\mathrm{K}$ initially to $294.86\,\mathrm{K}$ at $6\,\mathrm{min}$, $303.54\,\mathrm{K}$ at $12\,\mathrm{min}$, and $312.05\,\mathrm{K}$ at $60\,\mathrm{min}$, which already shows the trend in a clear way. Relative to the COMSOL steady-state value, the response reaches approximately 90% of the total temperature rise by $19\,\mathrm{min}$, 95% by $24\,\mathrm{min}$, and 99% by $32\,\mathrm{min}$, so basically most of the average heating is already there before the full 60 minute result. The final COMSOL value of $312.05\,\mathrm{K}$ differs from the analytical estimate by only $0.10\,\mathrm{K}$ ($0.03\,\%$), which is a very small error and because of that it shows excellent agreement between both methods.
This close agreement support the conclusion that:

- The σ_{eff} correction correctly maps the 3D trace geometry to the 2D model
- The Joule heating and convective boundary conditions are consistently implemented
- The total energy balance of the model is accurate

3.4 Sensitivity Analysis

The parametric sweeps performed in Sim 2 and Sim 2b serve as a de facto sensitivity analysis, showing how the minimum surface temperature responds to changes in the three main design variables.

Table 14: Sensitivity of minimum surface temperature to design parameters (Pd–Ag alloy, baseline: 14 traces, $12\,\mathrm{V}$).

Parameter change	ΔT_{min} (K)	Sensitivity
n_{traces}: 14 $\rightarrow$ 16 (+2 traces)	+1.40	$0.70\,\mathrm{K}$/trace
V_{in}: 12 $\rightarrow$ $14.5\,\mathrm{V}$ (+$2.5\,\mathrm{V}$)	+1.16	$0.46\,\mathrm{K\,V^{-1}}$
Material: Pd–Ag $\rightarrow$ Nichrome V	-1.49	—

The results confirm that **the number of traces is the most influential parameter**, followed by applied voltage. The choice of trace material has a strong effect through its conductivity: higher conductivity produces more Joule heating per unit length of trace.
The external heat transfer coefficient h_{ext} is also a sensitive parameter (as noted in the literature [9]), since it directly controls the rate of heat loss from the glass. This was systematically addressed in Sim 6 (Section 2.8), which swept h_{ext} alongside T_{amb} and T_{dew} across five realistic weather scenarios, confirming that the final design remains robust across all tested conditions.

4 Results and Discussion

4.1 Results

Trace Pattern Optimisation

With material, trace count, and voltage fixed at the baseline optimum (Pd–Ag alloy, 20 traces, $12\,\mathrm{V}$), the trace pattern was optimised without adding any cost. Two strategies were compared: a zoned layout (Sim 4, 6/8/6 distribution) and a continuous Chebyshev distribution with variable exponent p (Sim 5). The zoned layout showed only a small improvement time ($11\,\mathrm{min}$ defrost, $T_{\text{min,SS}} = 289.3\,\mathrm{K}$). The Chebyshev approach, which changes the trace pattern for different values of p, proved more effective: sweeping p identified an optimum at $p = 1.3$ and Table 15 compares all four layouts.

Table 15: Trace-pattern comparison (Pd–Ag alloy, 20 traces, $V_{\text{in}} = 12\,\mathrm{V}$).

Layout	Defrost time	$T_{\text{min,SS}}$ (K)	Gain vs. uniform
Uniform (baseline)	12 min	288.2	—
Zoned 6/8/6	11 min	289.3	-8%, $+1.1\,\mathrm{K}$
Chebyshev $p = 1.0$	8 min	292.6	-33%, $+4.4\,\mathrm{K}$
Chebyshev $p = 1.3$	**6 min**	**295.0**	$-$**50%**, $+6.8\,\mathrm{K}$

The Chebyshev $p = 1.3$ spacing halves the defrost time and raises the steady-state minimum temperature by $6.8\,\mathrm{K}$ relative to the base configuration, purely through geometric redistribution of the same 20 traces. The design also remains within the SAE J216 visibility limit with $4.25\,\%$ obscuration. Combining the material, electrical, and geometric optima gives the final design for the trace pattern. The $14.5\,\mathrm{V}$ Pd–Ag cases was not carried further, because Sim 2c showed they go above the $165\,\mathrm{MPa}$ glass limit and so it had to be removed. Sim 6 further confirmed that this configuration is robust five weather scenarios ranging from $-20\,^{\circ}\mathrm{C}$ to $10\,^{\circ}\mathrm{C}$ ambient and $10\,\mathrm{W\,m^{-2}\,K^{-1}}$ to $40\,\mathrm{W\,m^{-2}\,K^{-1}}$ for external convection. Every case reaches dew temperature within $8\,\mathrm{min}$ and is able to keep a steady-state margin of at least $5.4\,\mathrm{K}$ above its dew point. Therefore can be concluded that the final trace pattern configuration is the recommended outcome of this study.

4.2 Discussion

Strengths

this whole chapter seems unnecessary
The optimized trace pattern is able to achieve improved perfomance solely base on trace placement, at zero additional cost. The selected $12\,\mathrm{V}$ Pd–Ag design stays below the $165\,\mathrm{MPa}$ glass limit, while the $14.5\,\mathrm{V}$ Pd–Ag

cases do not. The highest stress is on the fixed side edges of the model, which is a bit conservative because in real life there is rubber around the glass and it can move a little bit too. The design operates at nominal $12\,\mathrm{V}$ battery voltage, meaning it works even with the engine off. The parametric approach (sweeping material, trace count, voltage, and spacing exponent) provides a comprehensive design map that allows easy trade-off analysis for different requirements, while the multiphysics coupling between Electric Currents, Heat Transfer, and Solid Mechanics captures the essential physics of the defroster operation. The mesh convergence study confirms that results are mesh-independent for the recommended configurations, and the boundary conditions (h_{ext}, h_{int}) are supported by published experimental data [9]. **The design is also robust across real-world weather conditions (Sim 6)**: all five scenarios — spanning ambient temperatures from $-20\,^{\circ}\mathrm{C}$ to $10\,^{\circ}\mathrm{C}$, external convection coefficients from $10\,\mathrm{W\,m^{-2}\,K^{-1}}$ to $40\,\mathrm{W\,m^{-2}\,K^{-1}}$, and physically realistic dew points — were defrosted within $8\,\mathrm{min}$, with a minimum steady-state margin of $5.4\,\mathrm{K}$ above the dew point (freezing rain scenario). Finally, the sequential solve strategy for the stress analysis overcomes the COMSOL 6.2 spatial frame limitation while still capturing the thermal stress state.

Limitations, Future Work, and Broader Implications

- **2D approximation:** The model uses a flat 2D geometry. A real rear window is curved, which affects heat flux distribution and stress concentrations. A 3D simulation with curved glass would be more realistic.
- **Fixed boundary conditions:** The fixed constraint at the glass edges overestimates the peak stress. Modelling the rubber seal as an elastic support would give more realistic stress values.
- **Constant heat transfer coefficients:** In reality, h_{ext} depends on vehicle speed (forced convection) and h_{int} depends on cabin HVAC settings. Sim 6 addressed the former by sweeping h_{ext} from $10\,\mathrm{W\,m^{-2}\,K^{-1}}$ to $40\,\mathrm{W\,m^{-2}\,K^{-1}}$ across five weather scenarios; the effect of varying h_{int} with HVAC state remains as future work.
- **Material property validation:** The EduPack values for Pd–Ag alloy are for bulk material. Screen-printed thick-film pastes may have different effective properties. Experimental validation with a physical prototype is recommended.
- **Contact resistance:** The model does not account for electrical contact resistance between traces and bus bars, which may reduce the effective heating power.

The Chebyshev spacing result has broader implications beyond this specific defroster. The key insight is that **non-uniform spatial distribution of heat sources**, optimised using a single continuous parameter, can be as effective as changing materials or operating conditions. This principle applies to any distributed heating or cooling problem where temperature uniformity is the objective.
In the context of automotive defrosters, the Chebyshev approach could be directly applied to front windshield defrosters (where driver visibility constraints are even more stringent) or to heated mirrors. The p-parameter provides a simple manufacturing specification: a single number that fully defines the trace layout for any given window height and trace count.
Furthermore, the observation that bulk-conductivity materials (Silver, Gold) overheat catastrophically in this geometry has a practical consequence: screen-printed thick-film pastes are not merely a cost-saving alternative to bulk metals — they are *functionally necessary* for the correct thermal behaviour. This finding is consistent with the industrial practice of using Pd–Ag and Ag paste formulations with controlled conductivity [8].
STILL DO: ELEMENT ORDER CONVERGENCE ANALYSIS — comparing linear vs. quadratic elements for the Joule heating/temperature solution to verify that the default element order is sufficient.
STILL DO: ADAPTIVE MESH REFINEMENT ANALYSIS — using COMSOL's adaptive mesh refinement to verify error control near the trace boundaries where temperature gradients are steepest.

5 Conclusion

In this report the design and optimisation of a car rear-window defroster using has been presented using coupled multiphysics finite element modelling in COMSOL Multiphysics 6.2. In order to complete the design of the heating trace pattern six simulations have been analyzed each with its own respective optimization part. Each stage narrowed the design space using the results of the previous one, producing a final configuration that satisfies the dew-point, defrost-time, stress, and obscuration requirements simultaneously.
Four main conclusions can be drawn from the study. First, the number of heating traces is by far the most influential design parameter as increasing the trace count from 12 to 20 at $12\,\mathrm{V}$ raises the minimum steady-state temperature T_{min} by more than $6\,\mathrm{K}$. In contrast raising the voltage from $12\,\mathrm{V}$ to $14.5\,\mathrm{V}$ at fourteen traces gives an increase of only $1.2\,\mathrm{K}$.

Second, out of all the materials the Pd–Ag alloy ($\sigma = 2.3 \times 10^6\,\mathrm{S\,m^{-1}}$) is a clear winner in terms of heat generation and distribution, reaching the dew point with fourteen
traces at $12\,\mathrm{V}$ where Nichrome V needs eighteen to twenty. However, bulk-conductivity materials such as silver and gold overheat too much in this geometry and cannot be used for the trace pattern optimization.
Third, once the COMSOL union result is interpreted as glass stress, the voltage choice changes a bit. The Pd–Ag cases at $12\,\mathrm{V}$ stays below the tempered-glass fracture strength of $165\,\mathrm{MPa}$ [7], but the Pd–Ag cases at $14.5\,\mathrm{V}$ goes above it and were therefore not selected anymore. The highest stresses are on the fixed model edges, which is conservative because a real rear window sits in rubber and can move a bit also. Fourth, geometry optimisation proves to be as meaningful as material or electrical optimisation. With the application of the Chebyshev trace spacing (Equation (8)) with exponent $p = 1.3$ is shown that the defrost time is halved from $12\,\mathrm{min}$ to $6\,\mathrm{min}$ and the T_{min} is raised by $6.8\,\mathrm{K}$ with zero additional cost which resulted to be the optimal design.
More importantly, the recommended design is not tuned to a single favourable operating temperature, to check for validity five weather scenarios are tested in Sim 6 — ranging ambient temperatures from $-20\,^{\circ}\mathrm{C}$ to $10\,^{\circ}\mathrm{C}$, external convection coefficients from $10\,\mathrm{W\,m^{-2}\,K^{-1}}$ to $40\,\mathrm{W\,m^{-2}\,K^{-1}}$ and physically realistic dew points Sebtain do you know this source? [1] — all defrost within $8\,\mathrm{min}$ with a minimum steady-state margin of $5.4\,\mathrm{K}$ above the dew point, confirming robustness across the expected operating envelope.
The simulation methodology itself with coupling Electric Currents, Heat Transfer, and Solid Mechanics through parametric sweeps over material properties, geometry, and operating conditions provides a systematic and reusable framework for optimization of trace pattern optimization. Furthermore, the result is supported by mesh convergence studies, back-of-envelope analytical estimates (agreeing to within 0.03%) and boundary conditions. To conclude the key insight is that the Chebyshev spacing rule offers a mathematically principled formula based on a single-parameter that can be applied to any rectangular heated-glass application. Future work should extend the model to three-dimensional glass, validate the simulation results against a physical prototype and investigate the effect of varying interior convection (h_{int}) under different car air conditioning states.

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